

# Problem Set 10: Sampling Distributions and Project Setup

SPEA V706 Math Camp | Module 10 · Friday, August 14

**Part A** is the algebra of estimators. **Part B** has you build a real project from an empty folder and then use it to watch the CLT happen in your own simulation.

## Part A. Concepts

### A1. The sampling distribution of a mean

A population has mean  $\mu = 20$  and standard deviation  $\sigma = 6$ . You draw an iid sample of size  $n = 36$  and compute  $\bar{Y}_n$ .

- What is  $E[\bar{Y}_n]$ ? Which property is this?
- What is  $\text{Var}(\bar{Y}_n)$ ?
- What is the standard error of  $\bar{Y}_n$ ?
- How large would  $n$  have to be to get the standard error down to 0.5?
- Is  $\bar{Y}_n$  Normally distributed here?

### A2. The rate

- Starting from  $\text{se}(\bar{Y}_n) = \sigma/\sqrt{n}$ , show algebraically that halving the standard error requires quadrupling  $n$ .
- A colleague has 1,000 observations and a standard error of 0.40. They want the standard error to be 0.10. How many observations do they need in total?
- What does this imply for the returns to collecting more data?

### A3. True or false

For each statement, say true or false and give one sentence of justification.

- The law of large numbers says that if a fair coin comes up heads ten times in a row, tails is now more likely on the next flip.
- The central limit theorem says that with a large enough sample, the variable  $Y$  itself becomes Normally distributed.
- The central limit theorem requires the population distribution to be Normal.

- d. If an estimator is unbiased, then the estimate from my particular sample is close to the truth.
- e. As  $n$  grows,  $\text{Var}(\bar{Y}_n)$  goes to zero.
- f. The standard deviation of  $Y$  and the standard error of  $\bar{Y}_n$  measure the same thing in different units.

#### A4. Unbiasedness, line by line

Show that  $E[\bar{Y}_n] = \mu$  when  $Y_1, \dots, Y_n$  are iid with mean  $\mu$ . Write out each step and name the expectation rule (E1, E2, or E3) that justifies it. Then answer: which part of the iid assumption did you use, and which part did you not?

#### A5. When the approximation is poor

The CLT holds in the limit, and “large enough  $n$ ” depends on the population.

- a. Give two features of a population distribution that would make  $n = 30$  too small for the Normal approximation to be any good.
- b. Give one feature of the *sampling scheme*, rather than the population, that can break the  $\sigma^2/n$  formula entirely.
- c. In the `wage1` data, the standard deviation of `wage` is about \$3.69 and there are 526 workers. Compute the standard error of the mean wage. Then say in one sentence what each of those two numbers describes.

## Part B. R

This part is graded on the folder, not just the answers. Build it properly.

### B1. Build a project from scratch

Create a new folder called `sampling_lab` and set it up the way deck **W5** describes.

- a. Create the RStudio project file (`sampling_lab.Rproj`) with File > New Project > New Directory.
- b. Create subfolders: `code/`, `outputs/figures/`, and `raw_data/`.
- c. Create a `.gitignore` that excludes `raw_data/` and `.Rproj.user/`.
- d. Create an empty `main.r` at the top level.
- e. Explain in one sentence why `here::here("code", "01_simulate.R")` is better than `setwd("/Users/yourname/Desktop/sampling_lab")`.

## B2. code/01\_simulate.R

Write a script that draws sampling distributions of the mean.

The population is Exponential with rate 1, so  $\mu = 1$  and  $\sigma = 1$ . For each of  $n = 5$ ,  $n = 30$ , and  $n = 200$ , draw 5,000 samples of that size and record the mean of each.

```
# code/01_simulate.R
pacman::p_load(data.table, here)
set.seed(20260814)

n_sims <- 5000
sizes  <- c(5, 30, 200)

# your code: for each size, 5000 sample means
# result: a data.table with columns n and xbar (15000 rows)

saveRDS(sims, here("outputs", "sampling_sims.rds"))
message("01_simulate.R complete")
```

- Fill in the simulation. `replicate()` or `matrix()` plus `rowMeans()` both work; the matrix version is much faster.
- Confirm your result has 15,000 rows and the columns you expect.

## B3. code/02\_analyze.R

- For each  $n$ , compute the mean of `xbar` across the 5,000 samples. Compare each to the true population mean of 1. What does the LLN predict?
- For each  $n$ , compute the standard deviation of `xbar`. Compare each to the theoretical  $\sigma/\sqrt{n} = 1/\sqrt{n}$ . Report the two side by side in a small table.
- Check the rate directly: divide the standard deviation at  $n = 5$  by the one at  $n = 200$ . The sample sizes differ by a factor of 40, so what ratio should you see? Does the simulation agree?
- Make one figure showing all three sampling distributions, save it to `outputs/figures/` with `ggsave()`.

## B4. Tie it together

- Write `main.r` so it sources both scripts in order using `here()`.
- Restart R (Session > Restart R), then run `main.r` from a clean session. Confirm it produces the saved figure with no errors and no objects left over from your interactive session.
- Delete `outputs/sampling_sims.rds` and the figure, then run `main.r` again. Everything should come back. If it does not, something in your pipeline depends on state you did not write down.

### B5. Read your own figure

- a. At which  $n$  does the sampling distribution still look visibly skewed?
  - b. At which  $n$  does it look Normal?
  - c. The population you sampled from is strongly right-skewed and nothing like a bell curve. Explain in two or three sentences why the sample means look Normal anyway, and what would have to be true about the population for this to fail.
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**Check yourself.** Without notes:

- 1. What is the difference between a standard deviation and a standard error?
- 2. Why does the sample mean have variance  $\sigma^2/n$  rather than  $\sigma^2$ ?
- 3. What does the CLT let you do that the LLN alone does not?